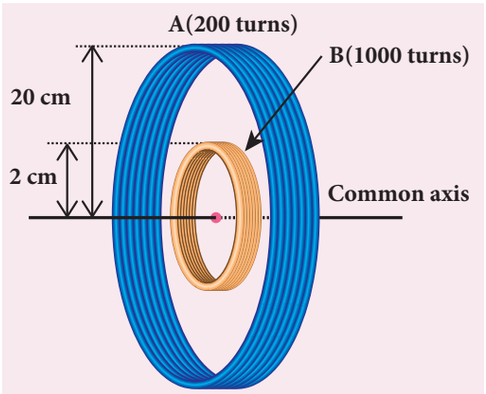
(ii) Induced emf in the second coil due to the rate of change of current in the first coil is

$$\varepsilon_2 = M\frac{di_1}{dt} = \frac{3\times10^{-3}\times12}{0.03}$$

$$\varepsilon_2 = 1.2\,\text{V}$$

## EXAMPLE 4.13

Consider two coplanar, co-axial circular coils *A* and *B* as shown in figure. The radius of coil *A* is 20 cm while that of coil *B* is 2 cm. The number of turns in coils *A* and *B* are 200 and 1000 respectively. Calculate the mutual inductance between the coils. If the current in coil *A* changes from 2 A to 6 A in 0.04 s, determine the induced emf in coil *B* and the rate of change of flux through the coil *B* at that instant.




### *Solution*

$N_A = 200$ turns; $N_B = 1000$ turns;
$r_A = 20\times10^{-2}$ m; $r_B = 2\times10^{-2}$ m;
$dt = 0.04$ s; $di_A = 6-2 = 4$ A

Let $i_A$ be the current flowing in coil *A*, then the magnetic field $B_A$ at the centre of the circular coil *A* is

$$B_A = \frac{\mu_\circ N_A i_A}{2r_A} = \frac{4\pi\times10^{-7}N_A i_A}{2r_A}$$
$$= \frac{10^{-7}\times2\times3.14\times200}{20\times10^{-2}}\times i_A$$
$$= 6.28\times10^{-4} i_A \ \text{Wb m}^{-2}$$

The magnetic flux linkage with coil *B* is

$$N_B\Phi_B = N_B B_A A_B$$
$$= 1000\times6.28\times10^{-4}\times i_A\times3.14\times\left(2\times10^{-2}\right)^2$$
$$= 7.89\times10^{-4} i_A \ \text{Wb turns}$$

The mutual inductance between the coils

$$M = \frac{N_B\Phi_B}{i_A} = 7.89\times10^{-4}\,\text{H}$$

Induced emf in coil *B* is

$$\varepsilon_B = -M\frac{di_A}{dt}$$
$$\varepsilon_B = \frac{7.89\times10^{-4}\times(6-2)}{0.04}\ \text{(magnitude only)}$$
$$\varepsilon_B = 78.9\,\text{mV}$$

The rate of change of magnetic flux of coil *B* is

$$\frac{d(N_B\Phi_B)}{dt} = \varepsilon_B = 78.9\ \text{m Wb s}^{-1}$$

## 4.4 METHODS OF PRODUCING INDUCED EMF

### 4.4.1 Introduction

Electromotive force is the characteristic of any energy source capable of driving electric charge around a circuit. We have already learnt that it is not actually a force. It is the work done in moving unit electric charge around the circuit. It is measured in J $C^{-1}$ or volt.

Some examples of energy source which provide emf are electrochemical cells, thermoelectric devices, solar cells and electrical generators. Of these, electrical



12th Physics_Vol-1_Unit 4_English Medium.indd 218 23-12-2022 17:57:22